GIFT IFIH · YOUNG BUILDERS PROGRAM · IFSC

AETHER-DERIVATIVES

Autonomous Margin & Collateral Intelligence for Cross-Border OTC Derivatives

"From contract interpretation to collateral decision — with deterministic financial computation and an auditable trail."

AI interprets. Deterministic engines calculate. Optimisation decides. Humans govern ambiguity.

THE PROPOSITION

Over-the-counter derivatives move on collateral. When volatility spikes, the desks that have to move it are reading legal annexes, rebuilding exposure in spreadsheets and reaching for cash because cash is easy.

AETHER deploys three agents over deterministic financial engines: one reads the Credit Support Annex, one turns exposure into a margin requirement, one selects the cheapest collateral that satisfies every contractual constraint — and every step leaves an auditable record.

- **Legal intelligence** — bespoke annexes, cited and confidence-scored
- **Deterministic risk** — no model ever produces a margin number
- **Cheapest-to-deliver** — constrained optimisation, not habit
- **Cross-border economics** — FX drag priced, not assumed

All market data, counterparties and legal documents in this prototype are SIMULATED. No transaction is executed, no market infrastructure is connected, no regulatory approval is claimed.

THE PROBLEM

Margin calls are settled by people reading legal documents.

$600 TRILLION

notional in global OTC derivatives markets

4 – 6 HOURS

to recalculate variation margin manually when volatility spikes

200+ PAGES

of ISDA Master Agreement and Credit Support Annex per counterparty

During volatility, human risk desks work from spreadsheets and manual PDF searches across those annexes.

Read the annex
threshold · MTA · haircuts

→

Rebuild exposure
netting set · mark to market

→

Compute the call
credit support amount

→

Choose collateral
eligibility · concentration

→

Instruct settlement
custodian · value date

→

Defend it later
audit · dispute

Late margin calls

trigger default notices and frozen trading lines.

Cash-first habit

posts the most expensive asset on the balance sheet.

No single record

leaves disputes to be argued from four systems.

These decisions sit across legal documents, risk engines, treasury spreadsheets and operations workflows. AETHER connects them.

Market scale and manual-effort figures as stated in the AETHER Master Playbook. Not measured by this prototype.

THE IDEA, WITHOUT THE JARGON

Think of it as an automated escrow desk.

Two parties trade. Prices move. Someone has to post security, and the contract decides how much and in what form.

01

THE CONTRACT AUDITOR

Legal-Clause Agent

Reads the agreement to establish the rules: which assets qualify as security, what minimum sizes apply, and what safety margin is taken on each.

`Threshold · Minimum Transfer Amount · eligible collateral · haircuts`

02

THE PIT BOSS

Risk-Engine Agent

Watches the position. When prices move, it works out exactly how much additional security is required right now, under those rules.

`Netting-set exposure · credit support amount · initial margin`

03

THE TREASURY MANAGER

Collateral-Optimizer Agent

Looks across everything available — cash, government bonds, corporate paper, gold — and picks the cheapest combination the contract actually permits.

`Eligibility · concentration caps · FX · funding cost`

Today all three roles are people, working in different systems, on different timelines. AETHER makes them one pipeline — and keeps a human in the loop exactly where the contract is ambiguous.

WHY THIS NEEDS MORE THAN A CHATBOT

A prompt wrapper cannot do this job.

LAYER	WHY A GENERIC LLM OR RAG FAILS	WHY AN AGENTIC SWARM SUCCEEDS
LEGAL EXTRACTION	Standard retrieval cannot correlate nested clauses — a Threshold modified by a credit-rating downgrade trigger sits in two paragraphs that never appear together.	The agent audits the annex systematically and emits a structured schema: typed values, each with a citation, each with a confidence, ambiguity routed to a human.
RISK ENGINE	Language models hallucinate arithmetic. They cannot be trusted with a nine-figure margin number, and no amount of prompting changes that.	The agent never calculates. It invokes audited deterministic functions through a typed tool boundary and the engine returns every figure.
COLLATERAL OPTIMISATION	Fixed rules cannot weigh cross-asset yields, haircuts, concentration caps and FX conversion simultaneously. Desks fall back on ‘post cash’.	The agent hands the problem to a linear program that solves it exactly — and returns the shadow price of every binding constraint.

AI interprets. Deterministic engines calculate.

Three agents. One orchestrator. One auditable decision.

AETHER ORCHESTRATOR

state, routing and hand-off between agents

LEGAL-CLAUSE AGENT

Interprets the contract

- Parses the ISDA Master Agreement and Credit Support Annex
- Extracts thresholds, minimum transfer amounts and valuation percentages
- Cites the source span for every value
  Flags ambiguity for human review

`OUT → structured contractual terms`

RISK-ENGINE AGENT

Quantifies the requirement

- Marks the netting set to a pinned market snapshot
- Applies the CSA waterfall: threshold, independent amount, MTA, rounding
- Computes initial margin alongside
  Never performs arithmetic itself

`OUT → the margin requirement`

COLLATERAL-OPTIMIZER AGENT

Decides what to deliver

- Screens inventory for eligibility under the annex's schedule
- Prices haircuts, FX and funding cost per unit actually delivered
- Solves for cheapest feasible mix
  Renders the settlement instruction

`OUT → the collateral decision`

DECISION RECORD → AUDIT CHAIN

Each agent's request and each engine's result is recorded separately.

THE WORKFLOW

One decision pipeline, from contract to collateral.

LEGAL DOCUMENT
ISDA Master Agreement + Credit Support Annex

LEGAL-CLAUSE AGENT — AGENT
reads, extracts, cites, scores confidence

HUMAN REVIEW — HUMAN
only where the contract is genuinely ambiguous

STRUCTURED TERMS
frozen — this version governs the calculation

RISK / MARGIN ENGINE — ENGINE
exposure, credit support amount, MTA, rounding

COLLATERAL OPTIMISER — ENGINE
cheapest feasible mix under every constraint

SETTLEMENT INSTRUCTION
structured message, pending human approval

AUDIT RECORD — CHAIN
who decided what, on what basis, in what order

The rule the pipeline is built on

A contractual value crosses from the model into the engine exactly once — through a validated schema, carrying a quote from the source document, past a human when confidence is low.

`After that point, no language model touches a number.`

WHAT EACH STAGE LEAVES BEHIND

Extraction	model, prompt version, confidence, citations
Human review	reviewer identity, decision, written rationale
Terms freeze	version number, document content hash
Each engine call	inputs, outputs, engine version, snapshot
Decision	action, amount, and why it was taken

Chained by hash, so altering one record breaks every record after it.

THE PRODUCT

Four capabilities — and where each one stands today.

01 **MULTI-MODAL LEGAL PARSER**

BUILT

Ingests the ISDA Master Agreement and Credit Support Annex and extracts threshold tiers, minimum transfer amounts and valuation percentages — every field cited to its source paragraph.

Native PDF and markdown today. Scanned documents without a text layer are refused, not guessed.

02 **DETERMINISTIC RISK ENGINE**

BUILT

Turns a netting set and a pinned market snapshot into a margin requirement: exposure, credit support amount, minimum transfer gate, directional rounding — plus initial margin alongside.

Two initial margin methods. Neither is ISDA SIMM: the weights are ours, uncertified, and documented as such.

03 **COLLATERAL OPTIMISATION**

BUILT

Evaluates the whole inventory — cash, sovereigns, corporate paper, gold — against haircut, eligibility, concentration and FX, and solves for the cheapest mix that satisfies every constraint.

A linear program, with the shadow price of every binding constraint returned alongside the allocation.

04 **SETTLEMENT MESSAGE GENERATION**

PARTIAL

Renders the approved allocation as the structured collateral messages a margin call actually uses, reconciled line by line to the delivery amount.

ISO 20022 and MT527 message SHAPE only. Nothing is transmitted — no network, no membership. Connectivity is roadmap.

Status is stated per capability rather than implied. What is built is demonstrable in the next two slides; what is not is named as roadmap.

THE COMMERCIAL THESIS

The expensive part of a margin call is what you deliver.

Two assets can satisfy the same call and cost very different amounts to give up.

THE RANKING THAT DECIDES THE ALLOCATION

cost / effective unit = funding spread ÷ effective valuation %

An asset with a low headline spread and a heavy haircut delivers less collateral per unit encumbered.

THE CONSTRAINTS IT MUST SATISFY SIMULTANEOUSLY

Eligibility
does the annex's schedule admit this asset at all?

Haircut
what percentage of market value actually counts?

Concentration
issuer, asset class and sector ceilings

Wrong-way risk
never the counterparty's own paper

Inventory
what is genuinely available to encumber

FX
non-base currency takes a further haircut

MODELLED reduction vs a cash-first baseline in one synthetic scenario. Not a guarantee, not annualised.

ON ONE VERIFIED RUN

Cash-first, as most desks post

USD 23,002.41

Optimised allocation

USD 5,496.52

MODELLED REDUCTION

76.10%

USD 17,505.89
on this one call

WHAT THE SOLVER ALSO RETURNS

The shadow price of every binding constraint — what relaxing it by one unit would save:

Coverage	6.80bp	Corporate cap	2.61bp
Gold cap	1.22bp	Single-issuer cap	1.09bp

Cross-border collateral has to be priced, not assumed.

An IFSC banking unit trades a USD-based agreement while funding rupee assets at home. Those two facts pull in opposite directions.

THE ADVANTAGE

Local paper is cheaper to fund at home. On headline spread it beats the alternative.

THE COST

It is not in the agreement's base currency, so the annex takes a further FX haircut on top of the schedule valuation.

AETHER resolves the tension by pricing it. On the verified run:

Headline funding spread	Schedule valuation	FX haircut on top	Effective valuation	Delivered cost	Cheapest alternative
9 bp	94.00%	× 92%	86.48%	10.41 bp	8.04 bp
cheaper than the 8 bp alternative	eligible collateral schedule	non-base currency provision	94% × 92%	cost per unit actually delivered	US Treasury bill

THE OPTIMISER'S DECISION

NOT SELECTED

The rupee asset was evaluated, priced, and declined — the Treasury bill delivered the same coverage for less.

OF WHICH, THE FX DRAG ALONE

0.83 bp — the price of posting cross-border, isolated from the schedule haircut

Elsewhere in the same inventory EUR cash is 28 bp on paper and 30.43 bp delivered — dearer than 30 bp USD cash.

The GIFT City case is that cross-border collateral is priced on delivered economics — not favoured because it is local.

WHY THIS CAN BE TRUSTED

Financial AI earns trust by refusing to guess.

DETERMINISTIC BY CONSTRUCTION

Every figure comes from a pure function of contract terms and a pinned market snapshot. No clock, no live feed, no randomness inside a calculation.

The same inputs produce byte-identical results, every time.

HUMAN AT THE AMBIGUITY

Where a clause is conditional — a threshold qualified by a ratings trigger the document cannot resolve — the pipeline stops and asks.

Freezing is refused server-side while any field awaits a decision.

EVIDENCE FOR EVERY STEP

Extraction, human confirmation, each engine call and the final decision are recorded in order and chained by hash.

Altering one record breaks every record after it.

AND WHAT AETHER DELIBERATELY DOES NOT CLAIM

ISDA SIMM	Public BCBS-IOSCO schedule plus our own uncertified sensitivity weights. Neither is SIMM.
SWIFT connectivity	Messages are generated in ISO 20022 shape and never transmitted. No network, no membership.
Real-time market data	Pinned snapshots. Published data can be ingested — but pinned before any calculation reads it.
Guaranteed savings	76.10% is modelled, in one synthetic scenario. Not annualised, not extrapolated.
Measured legal accuracy	In replay the cached extraction is hand-authored, so the score demonstrates the harness.
Production readiness	No authentication, no persistence, no multi-tenancy. A deliberately scoped MVP.

WHERE THIS GOES

From a verified prototype to institutional infrastructure.

NOW

VERIFIED PROTOTYPE

- Legal extraction with citations
- Deterministic margin waterfall
- Collateral optimisation with duals
- FX-aware cross-border pricing
- Human review gate
- Settlement message generation
- Hash-chained audit trail

NEXT

PRODUCTION INTEGRATION

- Live market-data feeds, pinned on ingest
- Real portfolio and trade ingestion
- Persistent, versioned agreement state
- Authentication and multi-user review
- Production databases and observability
- Custodian and tri-party connectivity
- Scanned-document ingestion

FUTURE

INSTITUTIONAL SCALE

- Multi-entity collateral pools
- Many netting sets in one optimisation
- Continuous intraday recalculation
- Multi-currency optimisation at scale
- Real settlement connectivity
- Enterprise multi-tenant deployment
- Dispute attribution

The decision engine scales conceptually — the linear program is indifferent to inventory size. Production infrastructure is the next engineering layer.

01 FINANCIALLY MEANINGFUL

Collateral selection treated as a constrained optimisation, with every binding constraint priced.

02 TECHNICALLY DEFENSIBLE

Agent orchestration over deterministic engines, a real solver, and a reproducible audit chain.

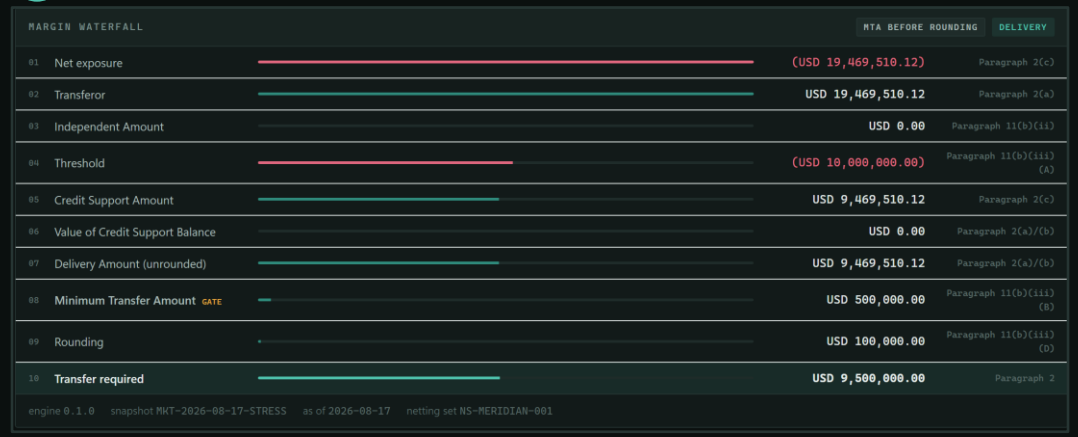
03 GIFT CITY RELEVANT

Cross-border collateral evaluated on delivered economics, FX drag included.

AETHER turns fragmented margin operations into one traceable decision pipeline.

The calculation, running.

01 Margin waterfall



MARGIN WATERFALL — MTA BEFORE ROUNDING — DELIVERY

01	Net exposure	(USD 19,469,510.12)	Paragraph 2(c)
02	Transferor	USD 19,469,510.12	Paragraph 2(a)
03	Independent Amount	USD 0.00	Paragraph 11(b)(ii)
04	Threshold	(USD 10,000,000.00)	Paragraph 11(b)(iii)(A)
05	Credit Support Amount	USD 9,469,510.12	Paragraph 2(c)
06	Value of Credit Support Balance	USD 0.00	Paragraph 2(a)/(b)
07	Delivery Amount (unrounded)	USD 9,469,510.12	Paragraph 2(a)/(b)
08	Minimum Transfer Amount GATE	USD 500,000.00	Paragraph 11(b)(iii)(B)
09	Rounding	USD 100,000.00	Paragraph 11(b)(iii)(D)
10	Transfer required	USD 9,500,000.00	Paragraph 2

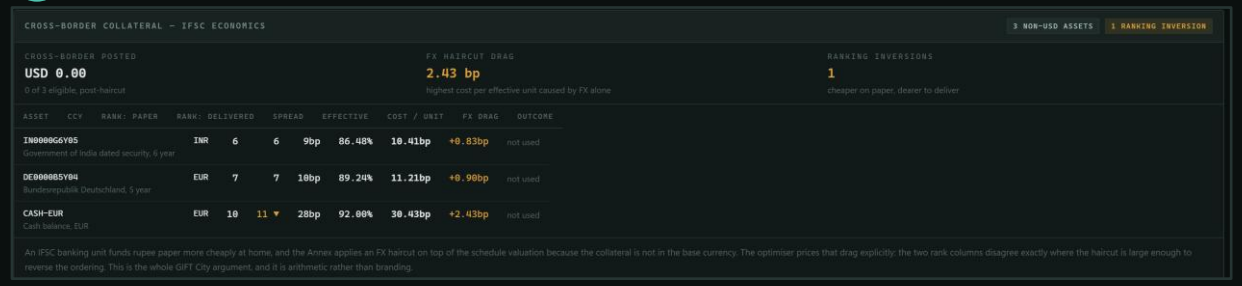
engine 0.1.0 snapshot MKT-2026-08-17-STRESS as of 2026-08-17 netting set NS-MERIDIAN-001

Exposure of (USD 19,469,510.12) less a USD 10m threshold gives a credit support amount of USD 9,469,510.12; the MTA gate and rounding produce USD 9,500,000.00.

ON THIS RUN

Cash-first baseline	USD 23,002.41
Optimised allocation	USD 5,496.52
Modelled reduction	76.10%
Binding constraints priced	4
Wrong-way asset	refused

03 Cross-border · IFSC economics



CROSS-BORDER COLLATERAL — IFSC ECONOMICS — 3 NON-USD ASSETS — 1 RANKING INVERSION

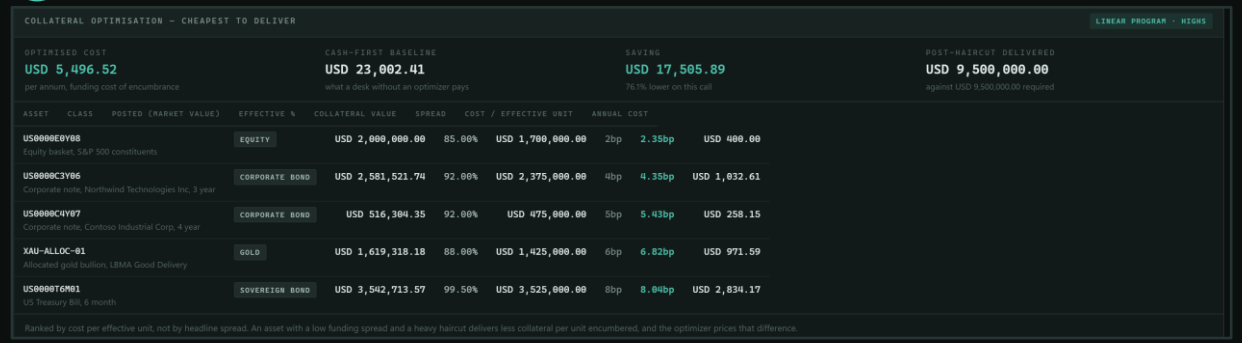
CROSS-BORDER POSTED: USD 0.00 — 0 of 3 eligible, post-haircut
FX HAIRCUT DRAG: 2.43 bp — highest cost per effective unit caused by FX alone
RANKING INVERSIONS: 1 — cheaper on paper, dearer to deliver

ASSET	CCY	RANK: PAPER	RANK: DELIVERED	SPREAD	EFFECTIVE	COST / UNIT	FX DRAG	OUTCOME
IN0000G6Y85 Government of India dated security, 6 year	INR	6	6	9bp	86.48%	10.41bp	+0.83bp	not used
DE0000B5Y84 Bundesrepublik Deutschland, 5 year	EUR	7	7	10bp	89.24%	11.21bp	+0.90bp	not used
CASH-EUR Cash balance, EUR	EUR	10	11 ▼	28bp	92.00%	30.43bp	+2.43bp	not used

An IFSC banking unit funds rupee paper more cheaply at home, and the Annex applies an FX haircut on top of the schedule valuation because the collateral is not in the base currency. The optimiser prices that drag explicitly: the two rank columns disagree exactly where the haircut is large enough to reverse the ordering. This is the whole GIFT City argument, and it is arithmetic rather than branding.

Every non-USD asset with its paper rank beside its delivered rank, FX drag isolated. The rupee asset is priced at 10.41bp and declined.

02 Collateral optimisation · cheapest to deliver



COLLATERAL OPTIMISATION — CHEAPEST TO DELIVER — LINEAR PROGRAM · HIGHS

OPTIMISED COST: USD 5,496.52 — per annum, funding cost of encumbrance
CASH-FIRST BASELINE: USD 23,002.41 — what a desk without an optimizer pays
SAVING: USD 17,505.89 — 76.1% lower on this call
POST-HAIRCUT DELIVERED: USD 9,500,000.00 — against USD 9,500,000.00 required

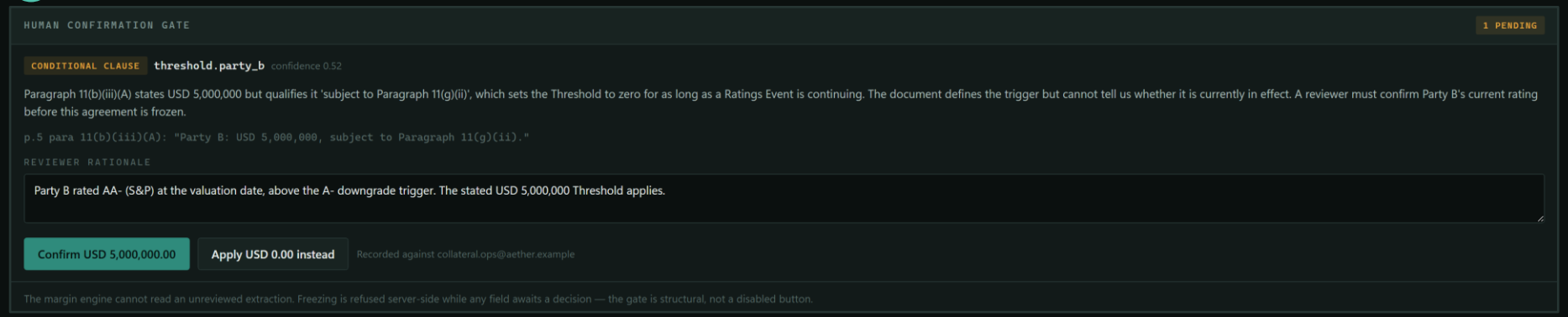
ASSET	CLASS	POSTED (MARKET VALUE)	EFFECTIVE %	COLLATERAL VALUE	SPREAD	COST / EFFECTIVE UNIT	ANNUAL COST
US0000E0Y08 Equity basket, S&P 500 constituents	EQUITY	USD 2,000,000.00	85.00%	USD 1,700,000.00	2bp	2.35bp	USD 400.00
US0000C3Y06 Corporate note, Northwind Technologies Inc, 3 year	CORPORATE BOND	USD 2,581,521.74	92.00%	USD 2,375,000.00	4bp	4.35bp	USD 1,032.61
US0000C4Y07 Corporate note, Contoso Industrial Corp, 4 year	CORPORATE BOND	USD 516,304.35	92.00%	USD 475,000.00	5bp	5.43bp	USD 258.15
XAU-ALLOC-01 Allocated gold bullion, LBMA Good Delivery	GOLD	USD 1,619,318.18	88.00%	USD 1,425,000.00	6bp	6.82bp	USD 971.59
US0000T6M01 US Treasury Bill, 6 month	SOVEREIGN BOND	USD 3,542,713.57	99.50%	USD 3,525,000.00	8bp	8.04bp	USD 2,834.17

Ranked by cost per effective unit, not by headline spread. An asset with a low funding spread and a heavy haircut delivers less collateral per unit encumbered, and the optimizer prices that difference.

Five assets, ranked by cost per effective unit rather than headline spread.

The governance, running.

04 Human confirmation gate · the pipeline stops



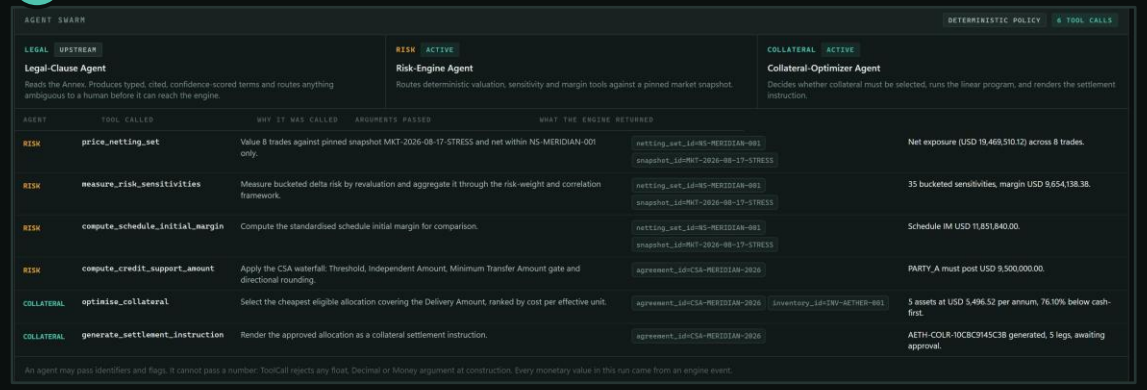
FLAGGED FIELD
`threshold.party_b`

CONFIDENCE
0.52 below the 0.85 gate

FREEZE
HTTP 409

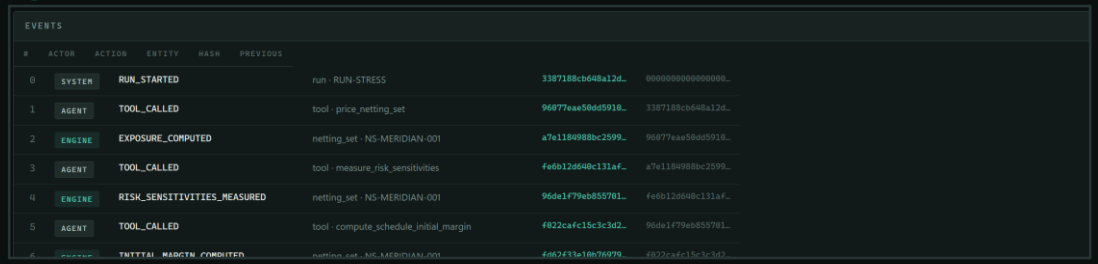
A conditional clause: the threshold is stated, then qualified by a ratings trigger the document cannot resolve. The agent does not guess — it flags the field, and the server refuses to freeze the agreement until a human decides.

05 Agent swarm · the tool boundary



Every argument an agent passes is an identifier — never a number.

06 Audit chain · 15 events, verified



Agent decision, then engine computation, alternating.

406	tests	100%	reproducible	0	network calls